

## Assignment (Linked) List Theory

1. What is Linked lists? What type of memory allocation is referred for Linked lists? what is traversal in linked lists?
2. Describe what is Node in link list? And name the types of Linked Lists and Explain each?
3. What will you prefer to use a singly or a doubly linked lists for traversing through a list of elements?
4. Explain the application of linked list with example in detail.
5. Mention the steps to insert data at the starting of a singly linked list? How to reverse a singly linked list?
6. Mention what is the difference between Linear Array and Linked List?
7. How to remove loops in a linked list (or) what are fast and slow pointers used for?
8. Compare Linked lists and Dynamic Arrays.
9. How many pointers are required to implement a simple Linked list? Justify the answer.
10. How to Delete Alternate Nodes Of A Linked List?
11. Given a single linked list L1 which is sorted in ascending order , and another single linked list L2 which is not sorted. Write a function to print the elements of second list according to the first list . For Example if the first list is 1->2->5->7->8->, then the function should print the 1<sup>st</sup> ,2<sup>nd</sup> ,5<sup>th</sup> ,7<sup>th</sup> ,8<sup>th</sup> element of the second list.
12. Write a function to insert a node just before and just after a node pointed to by a pointer p, without using the pointer start.
13. write a function to remove duplicates from a sorted linked list.
14. write a function to delete the alternate node (even numbered node) from a single linked list. For Example if the list is 1->2->3->4->5->6->7 then the resulting list should be 1->3->5->7.
15. Write a function to find out whether a single linked list is NULL terminated or contains a cycle /loop. If the list contain a cycle find the length of the cycle and length of the whole list. Find the node that causes the cycle i.e. the node at which the cycle starts. The node is pointed by two different nodes of list . Remove the cycle from the list and make it NULL terminated.
16. Create a double linked list in which info part of each mode contains a digit of a given number . the digits should be stored in reverse order i.e. the least significant digit should be stored in first node and most significant digit in last node. If the number is 5468132 then the linked list should be 2->3->1->8->6->4->5->. Write a function to add two numbers represented by linked list.